

Subject :- Cyber Security

Practical 5

Aim:- Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CODE:-

Diffie-Hellman Key Exchange

```
print("=====")
print("    DIFFIE-HELLMAN KEY EXCHANGE")
print("=====")
```

Public values

```
p = int(input("\nEnter a prime number (p): "))
g = int(input("Enter a generator (g): "))
```

Private keys

```
a = int(input("Enter Alice's private key: "))
b = int(input("Enter Bob's private key: "))
```

Calculate public keys

```
A = pow(g, a, p)
B = pow(g, b, p)
```

```
print("\n--- Public Key Exchange ---")
print("Alice's public key:", A)
print("Bob's public key:", B)
```

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Calculate shared secret keys

```
alice_shared_key = pow(B, a, p)
```

```
bob_shared_key = pow(A, b, p)
```

```
print("\n--- Shared Secret Keys ---")
```

```
print("Alice's shared key:", alice_shared_key)
```

```
print("Bob's shared key:", bob_shared_key)
```

Verify

```
if alice_shared_key == bob_shared_key:
```

```
    print("\nKey Exchange Successful!")
```

```
    print("Both Alice and Bob have the same shared secret key.")
```

```
else:
```

```
    print("\nKey Exchange Failed!")
```

output:-



```
IDLE Shell 3.13.1*
File Edit Shell Debug Options Window Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec 3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> == RESTART: C:/Users/Trupti/OneDrive/Desktop/RSA-Practical-2/diffie_hellman.py ==
=====
DIFFIE-HELLMAN KEY EXCHANGE
=====
Enter a prime number (p): 25
Enter a generator (g): 6
Enter Alice's private key: 7
Enter Bob's private key: 78

--- Public Key Exchange ---
Alice's public key: 11
Bob's public key: 16

--- Shared Secret Keys ---
Alice's shared key: 6
Bob's shared key: 6

Key Exchange Successful!
Both Alice and Bob have the same shared secret key.
>>> == RESTART: C:/Users/Trupti/OneDrive/Desktop/RSA-Practical-2/diffie_hellman.py ==
=====
DIFFIE-HELLMAN KEY EXCHANGE
=====
```

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